

GURU

Guru AI - SEE 2082 (2026) Model Solutions

Compulsory Mathematics (RE-1031'BP')

Ans 1. (Sets)

a. Let F and V be the sets of students playing football and volleyball respectively. Number of students playing both games in cardinality notation is $n(F \cap V) = 20$.

b. Total students $n(U) = 120$, neither game $n(\bar{F} \cup \bar{V}) = 10$. Ratio of $n_0(F) : n_0(V) = 2:1$.

c. Let $n_0(V) = x$. Then $n_0(F) = 2x$.

Formula: $n(U) = n_0(F) + n_0(V) + n(F \cap V) + n(\bar{F} \cup \bar{V})$

$$120 = 2x + x + 20 + 10 \Rightarrow 3x = 90 \Rightarrow x = 30$$

Number of students who like to play football $n(F) = n_0(F) + n(F \cap V) = 2(30) + 20 = 80$.

d. Number of students playing volleyball $n(V) = n_0(V) + n(F \cap V) = 30 + 20 = 50$.

Difference = $80 - 50 = 30$. Percentage less = $(30 / 80) \times 100\% = 37.5\%$. Thus, it is 37.5% less.

Ans 2. (Compound Interest)

Given: Rate $R = 10\%$, $CA_2 - CA_1 = \text{Rs. } 47,250$.

a. We know, $CA = P(1 + R/100)^T$

$$P(1.10)^2 - P(1.10)^1 = 47250 \Rightarrow P(1.21 - 1.10) = 47250 \Rightarrow 0.11P = 47250$$

Principal (P) = Rs. 4,29,545.45.

b. Semi-annual CI for 1 yr = $P(1 + 0.10/2)^2 - P = P(1.05)^2 - P = 0.1025P$

Quarterly CI for 1 yr = $P(1 + 0.10/4)^4 - P = P(1.025)^4 - P \approx 0.10381289P$

Difference = $(0.10381289 - 0.1025) \times 429545.45 = 0.00131289 \times 429545.45 = \text{Rs. } 563.95$.

Ans 3. (Depreciation)

a. Compound depreciation refers to the continuous reduction in the value of an asset over time at a certain percentage rate, calculated on the progressively decreasing book value.

b. Depreciation in first year = 20% of 50,000 = $(20/100) \times 50,000 = \text{Rs. } 10,000$.

c. Half of current price = $50000 / 2 = \text{Rs. } 25,000$.

Price after 3 years $P_3 = P(1 - R/100)^3 = 50000(1 - 0.20)^3 = 50000(0.8)^3 = 50000 \times 0.512 = \text{Rs. } 25,600$.

Since $\text{Rs. } 25,600 > \text{Rs. } 25,000$, it will NOT be possible to buy the machine at half its current price after 3 years.

Ans 4. (Exchange Rates)

a. The buying rate is used when exchanging US dollars into Nepalese rupees (since the bank is buying the dollars).

b. Nepalese rupees obtained = $3,000 \times 142.48 = \text{NRs. } 4,27,440$.

c. Devaluation of 0.5% means foreign currency becomes more expensive.

New Buying Rate = $142.48 + 0.5\% = 142.48 \times 1.005 = \text{NRs. } 143.1924$.

New Selling Rate = $143.08 + 0.5\% = 143.08 \times 1.005 = \text{NRs. } 143.7954$. (Difference is 0.603).

d. Gain for tourist = $(\text{New Buying Rate} - \text{Old Buying Rate}) \times 3000 = (143.1924 - 142.48) \times 3000 = 0.7124 \times 3000 = \text{NRs. } 2137.20$.

Ans 5. (Mensuration - Pyramid)

a. In the formula $V = 1/3 a^2 h$, 'a' denotes the length of the side of the square base.

b. $960 = 1/3 a^2 (5) \Rightarrow a^2 = 576 \Rightarrow a = 24 \text{ cm}$.

Slant height $l = \sqrt{(h^2 + (a/2)^2)} = \sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = 13 \text{ cm}$.

Total Surface Area (TSA) = $a^2 + 2al = 576 + 2(24)(13) = 576 + 624 = 1200 \text{ cm}^2$.

c. Area of base = $a^2 = 576 \text{ cm}^2$. Area of one triangular surface = $1/2 a l = 1/2(24)(13) = 156 \text{ cm}^2$.

Comparing them: The area of the base is $576 / 156 = 48/13 \approx 3.69$ times larger than the area of a single triangular surface.

Ans 6. (Cylinder & Hemisphere)

a. There are exactly 3 total external surfaces in the given combined solid (1 flat circular base, 1 curved surface of cylinder, 1 curved surface of hemisphere).

b. Radius $r = 14 / 2 = 7$ cm. Height of cylindrical part $h = 27 - r = 27 - 7 = 20$ cm.

c. Volume = Volume of Cylinder + Volume of Hemisphere

$$V = \pi r^2 h + \frac{2}{3} \pi r^3 = (22/7 \times 49 \times 20) + (\frac{2}{3} \times 22/7 \times 343)$$

$$V = 3080 + 718.67 = 3798.67 \text{ cm}^3.$$

Ans 7. (Paving Stones)

a. Area of parking = $16 \text{ m} \times 10 \text{ m} = 160 \text{ m}^2$.

Area of one stone = $0.25 \text{ m} \times 0.25 \text{ m} = 0.0625 \text{ m}^2$.

Total stones required = $160 / 0.0625 = 2560$ stones.

b. Cost of stones = $2560 \times \text{Rs. } 20 = \text{Rs. } 51,200$.

Labour cost = $3 \text{ workers} \times 2 \text{ days} \times \text{Rs. } 1800 = \text{Rs. } 10,800$.

Total estimated cost = $51200 + 10800 = \text{Rs. } 62,000$.

Ans 8. (Arithmetic / Geometric Series)

a. The sum of first n terms is $S_n = n/2 [2a + (n-1)d]$.

b. $S_5 = 50$. So, $50 = 5/2 [2(12) + (5-1)d] \Rightarrow 20 = 24 + 4d \Rightarrow 4d = -4 \Rightarrow d = -1$.

c. First three terms are 12, $(12-1)$, $(11-1) \Rightarrow 12, 11, 10$.

Subtracting 5 and 7 from the 2nd and 3rd terms respectively yields: 12, $(11-5)$, $(10-7) \Rightarrow 12, 6, 3$.

Ratio of consecutive terms: $6/12 = 1/2$ and $3/6 = 1/2$. Since the common ratio is equal, they form a geometric series.

Ans 9. (Quadratic Equations)

a. A quadratic equation is a polynomial equation of the second degree, generally represented as $ax^2 + bx + c = 0$, where $a \neq 0$.

b. Let the son's age in 2075 B.S. be X . Father's age = $6X$.

In 2080 B.S. (5 years later), son's age = $X + 5$, father's age = $6X + 5$.

Condition: $(X + 5)(6X + 5) = 350 \Rightarrow 6X^2 + 35X - 325 = 0$.

c. Solving: $6X^2 + 65X - 30X - 325 = 0 \Rightarrow X(6X + 65) - 5(6X + 65) = 0 \Rightarrow (X - 5)(6X + 65) = 0$.

Since age cannot be negative, $X = 5$. The son was 5 years old in 2075 B.S.

Birth year of the son = $2075 - 5 = 2070$ B.S.

Ans 10. (Algebra)

a. $1/(x-y) - (x+y)/(x^2-y^2) = 1/(x-y) - (x+y)/((x-y)(x+y)) = 1/(x-y) - 1/(x-y) = 0$.

b. $25^X - 6(5^{X+1}) + 125 = 0 \Rightarrow (5^X)^2 - 30(5^X) + 125 = 0$.

Let $y = 5^X$. Then $y^2 - 30y + 125 = 0 \Rightarrow (y - 25)(y - 5) = 0$.

If $y = 25 \Rightarrow 5^X = 5^2 \Rightarrow X = 2$.

If $y = 5 \Rightarrow 5^X = 5^1 \Rightarrow X = 1$.

Ans 11. (Geometry - Triangle Area)

a. The area of ΔPRS and ΔPRT are equal (because they stand on the same base PR and lie between the same parallel lines PR and ST).

b. Area of $\Delta PQT = \text{Area}(\Delta PQR) + \text{Area}(\Delta PRT)$.

Area of Quad $PQRS = \text{Area}(\Delta PQR) + \text{Area}(\Delta PRS)$.

Since $\text{Area}(\Delta PRT) = \text{Area}(\Delta PRS)$, it follows that $\text{Area}(\Delta PQT) = \text{Area of Quad } PQRS$.

c. From above, $\text{Area}(\Delta PRS) = \text{Area}(\Delta PRT)$.

Subtracting $\text{Area}(\Delta PRO)$ from both sides: $\text{Area}(\Delta PRS) - \text{Area}(\Delta PRO) = \text{Area}(\Delta PRT) - \text{Area}(\Delta PRO)$

Therefore, $\text{Area of } \Delta POS = \text{Area of } \Delta ROT$.

Ans 12. (Circle Geometry)

a. Relation: $\angle LOM = 2 \times \angle LNM$ (The central angle is twice the inscribed angle standing on the same arc LM).

b. In a circle, parallel chords intercept congruent arcs between them. Since $KL \parallel NM$, $\text{Arc } KN \cong \text{Arc } LM$.

Because equal arcs subtend equal chords, it is proved that $KN = LM$.

c. Experimental Verification Step: Draw two circles of radii at least 3 cm, construct cyclic quadrilateral $KLMN$, measure $\angle KLM$ and $\angle MNK$ and verify that their sum is 180° (Opposite angles of a cyclic quadrilateral are supplementary).

Ans 13. (Construction)

a. (Visual Construction Step) Draw base $BC = 5.5$ cm. At B , construct $\angle B = 75^\circ$. Mark A such that $AB = 4.5$ cm. Join AC to form $\triangle ABC$. Then, construct a parallelogram $FCD E$ with base equal to half of BC (or equivalent altitude) sharing parallel lines, ensuring $\angle EFC = 60^\circ$ to equal the area.

b. The area of a triangle and a parallelogram between the same parallel lines are equal when the base of the parallelogram is half the base of the triangle.

Ans 14. (Trigonometry)

a. Angle of elevation is the angle formed by the line of sight with the horizontal plane when an observer looks upward at an object.

b. (Visualizing Diagram) Right triangle ABC , where AB is the house height, $BC = 10$ m is the base, and $\angle ACB = 60^\circ$.

c. Height $AB = BC \times \tan(60^\circ) = 10 \times \sqrt{3} = 17.32$ m (or $10\sqrt{3}$ m).

d. Moving 20 m further away makes the total base $= 10 + 20 = 30$ m.

$\tan(\theta) = (10\sqrt{3}) / 30 = \sqrt{3} / 3 = 1 / \sqrt{3} \Rightarrow \theta = 30^\circ$.

Ans 15. (Statistics)

a. Formula for Mode $M_o = L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times i$.

b. Maximum frequency is 7, so the modal class is 50-60.

$$L=50, f_1=7, f_0=6, f_2=3, i=10.$$

$$M_o = 50 + [(7 - 6) / (14 - 6 - 3)] \times 10 = 50 + (1 / 5) \times 10 = 52.$$

c. Total $N = 25$. Median position $= N/2 = 12.5$. CF values: 3, 7, 13, 20, 23, 25. The CF just greater than 12.5 is 13, making the median class 40-50.

$$M_d = L + [(N/2 - c.f.) / f] \times i = 40 + [(12.5 - 7) / 6] \times 10 = 40 + 55/6 = 40 + 9.17 = 49.17.$$

d. Median class is 40-50. Total people below this class $= 3 + 4 = 7$. Total people above this class $= 7 + 3 + 2 = 12$. The number of people above the median class (12) is greater than those below it (7).

Ans 16. (Probability)

a. The multiplication law of probability for independent events is: $P(A \cap B) = P(A) \times P(B)$.

b. (Tree Diagram Representation - Without Replacement)

• 1st Draw: $P(R) = 2/5$, $P(W) = 3/5$

• 2nd Draw (if 1st was R): $P(R) = 1/4$, $P(W) = 3/4$

• 2nd Draw (if 1st was W): $P(R) = 2/4$, $P(W) = 2/4$

c. Probability of getting both red: $P(RR) = (2/5) \times (1/4) = 2/20 = 1/10$.

d. $P(\text{Same color}) = P(RR) + P(WW) = (2/5)(1/4) + (3/5)(2/4) = 2/20 + 6/20 = 8/20 = 2/5$.

$P(\text{Different color}) = P(RW) + P(WR) = (2/5)(3/4) + (3/5)(2/4) = 6/20 + 6/20 = 12/20 = 3/5$.

Comparing them, getting both balls of different colors (3/5) has a higher probability than getting both of the same color (2/5).